

ARUP

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Fax To: GOURGOUBES

Fax No: 010 33 6773 3065

Attn: HUMBERT

cc: ALEX @ RFR

From: ANDY SEDGWICK

Subject: T. P. L.

Direct Dialling:
Fax: 071-465 3743

Date: 9.5.91

Ref/Job no:

Total Pages:
(including this page) 7

If you have not received all the pages listed please phone the sender

Message

HUMBERT,

PETER, LORRAINE & I HAD A LONG DISCUSSION
YESTERDAY EVENING ABOUT THE DESIGN OF THE
MIRRORS FOR THE NEXT FULL MOON. IN THIS FAX
I HOPE TO SET DOWN MOST OF THE IMPORTANT
DESIGN ISSUES AND I WOULD LIKE YOU TO TELL
US IF THERE IS ANYTHING YOU ARE UNHAPPY WITH
OR THINK WILL BE IMPRACTICAL.

① THE MIRROR "SHELLS"

THE METHOD USED AT THE END OF APRIL SEEMED TO
GENERATE THE CORRECT OPTICAL SHAPE BUT WAS TOO
HEAVY. THE FOLLOWING IS THEREFORE PROPOSED:

- a) REDUCE TIMBER SIZE FROM 3CM x 3CM TO 1½CM x 1½CM
- b) KEEP 'BEAMS' AT SAME SPACING
AS BEFORE

can you buy
this?

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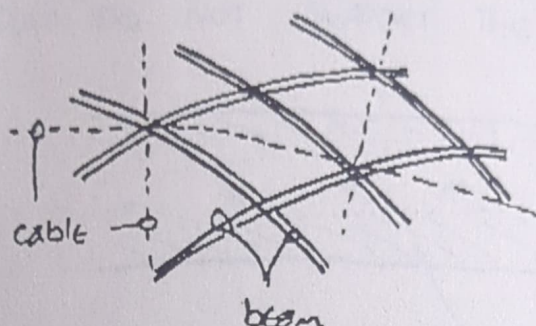
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- c) USE CABLES RUNNING DIAGONALLY TO HELP STIFFEN SHELL



THE CABLES CAN BE 'WOVEN' THROUGH THE TIMBER, AND FIXED IN A FEW PLACES TO THE WOOD.

- d) USE THINNER PLYWOOD (IF POSSIBLE)

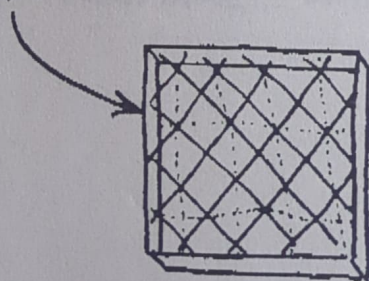
WITH THE ABOVE CABLES IN PLACE THE STRUCTURAL PROPERTIES OF THE PLYWOOD ARE NOT SO IMPORTANT - ITS MAIN PURPOSE IS TO ACT AS A SUPPORT FOR THE MIRROR SURFACE.

CAN YOU GET VERY THIN PLYWOOD? (SAY 3mm THICK?)

THE ABOVE MEASURES SHOULD MAKE THE MIRROR SHELLS A LOT LIGHTER, BUT STILL STIFF.... (WITH A FRAME)

2) SUPPORTING THE "SHELLS":

WE SHOULD TRY TO REDUCE THE WEIGHT OF THE EDGE STRUCTURE THAT ACTS AS A FRAME TO THE SHELL:



Job Title

Member/Location

Drg Ref

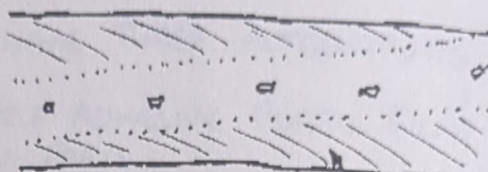
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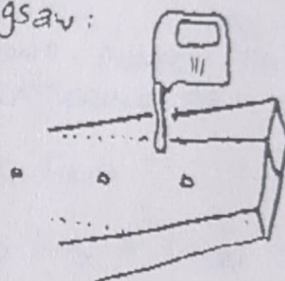
THIS CAN BE DONE BY TRIMMING AWAY PARTS OF THE FRAME THAT DO NOT SUPPORT THE SHELL WITH A JIGSAW :

original
frame
beam

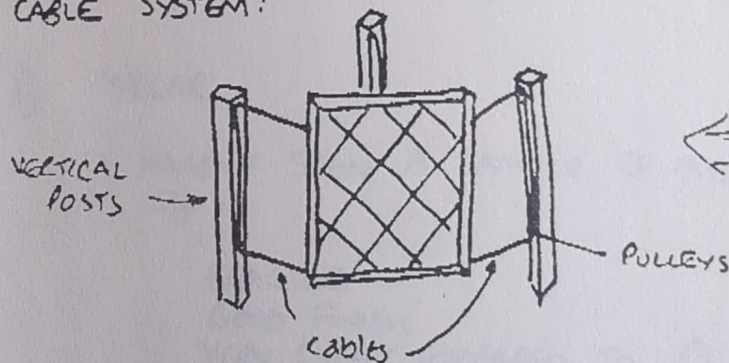


positions
where shell
meets frame

portion cut off with
jigsaw:



TO SUPPORT THE FRAME AND SHELL PETER SUGGESTS USING A CABLE SYSTEM:



SOMETHING
LIKE THIS

WE ARE STILL THINKING ABOUT THIS TO ENSURE THAT ALL REQUIRED MIRROR MOVEMENTS CAN BE ACHIEVED, BUT IT LOOKS MORE PROMISING THAN TRYING TO SUPPORT EVERYTHING WITH TIMBER FROM BELOW.

③ THE MIRRORS THEMSELVES :

WE SEEM TO HAVE THREE ALTERNATIVES :

- 1) 'JAL' SELF ADHESIVE PLASTIC FILM LIKE ALEX BOUGHT IN PARIS

THIS SEEMS VERY STRONG, BUT IT IS QUITE EXPENSIVE

IT DOESN'T HAVE A PERFECT FINISH; IT HAS A SLIGHT TEXTURE TO IT, BUT I DON'T EXPECT THIS WILL LEAD TO A SIGNIFICANT SCATTERING OF LIGHT

- 2) '3M' SELF ADHESIVE PLASTIC FILM

THIS IS QUITE STRONG, AND HAS A GOOD SMOOTH FINISH.

UNFORTUNATELY IT IS ABOUT TWICE AS EXPENSIVE AS 1) ABOVE

- 3) MYLAR

I HAVEN'T SEEN A SAMPLE OF MYLAR, BUT BELIEVE IT IS :

FRAGILE

GOOD FINISH

VERY CHEAP COMPARED TO 1) & 2)

WHAT DO YOU THINK ? IT WOULD BE GOOD TO USE THE MYLAR TO SAVE MONEY, BUT ~~IS~~ IS IT TOO FRAGILE TO BE PRACTICAL ?

④ WHAT TO MAKE THIS MONTH?

WE SUGGEST THAT WE TRY TO DESIGN AND MAKE THIS MONTH:

3 LARGE 'BACK OF HOUSE' REFLECTORS

SUGGEST 5M HIGH, 3M WIDE

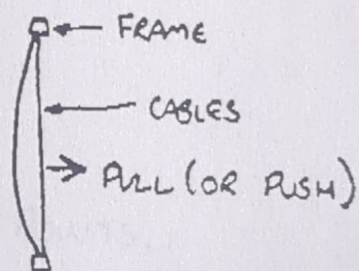
1 FIXED FOCUS SPOTLIGHT

SUGGEST 3M DIAMETER

1 VARIABLE FOCUS SPOTLIGHT

SUGGEST 3M DIAMETER

→ FOR THIS WE ARE THINKING OF 'ARCHERS BOW' PRINCIPLE TO CHANGE SHAPE:



WE NEED TO AGREE ~~SOON~~ SOON THE POSITIONS WHERE THESE CAN BE MOUNTED IN THE THEATRE AS THEY AFFECT THEIR DESIGN:

I HAVE SUGGESTED SOME APPROXIMATE POSITIONS ON THE ATTACHED SKETCH - PLEASE ADVISE WHAT IS POSSIBLE IN THEATRE CURRENTLY, AND GIVE ME DIMENSIONED LOCATIONS AS ACCURATELY AS YOU CAN.

Job Title

Member Location

Org Ref

Made by

Date

Chd.

⑤ SCHEDULE

WE HOPE TO BE ABLE TO SEND YOU THE DESIGNS FOR THE BIG 'BACK OF HOUSE' REFLECTORS BY THE MIDDLE OF NEXT WEEK.

THE DESIGNS FOR THE SUPPORTING STRUCTURES AND THE SPOTLIGHTS WILL FOLLOW A FEW DAYS LATER.

I WOULD LIKE TO COME TO GOURGOUBES FOR THE WEEKEND BEFORE THE FULL MOON TO HELP OUT COULD I STAY AT GOURGOUBES? I WOULD PROBABLY ARRIVE ON FRIDAY 24TH AND LEAVE MONDAY 27TH. WHAT DO YOU THINK?

A STUDENT FROM PENN CALLED THOR THORS HAS BEEN TO SEE US AND WOULD LIKE TO ALSO HELP OUT AT GOURGOUBES DURING THE NEXT FULL MOON. I'M GOING TO SPEAK TO HIM AGAIN ON MONDAY ... WOULD YOU LIKE HIM TO HELP?

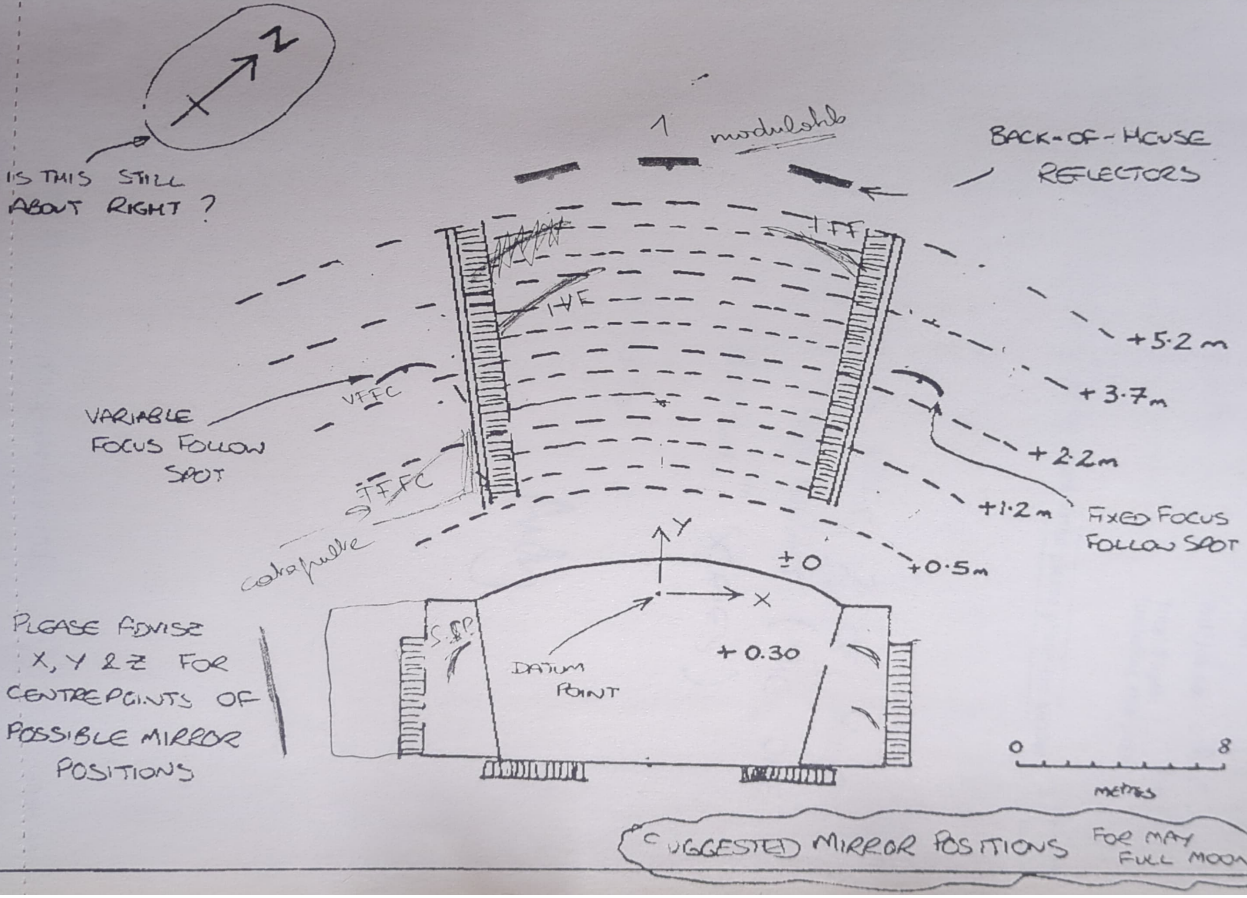
I THINK THEY ARE THE MAIN POINTS.

PLEASE LET ME KNOW IF THERE IS ANY THING YOU DISAGREE WITH IN WHAT WE PROPOSE, AND LET ~~ME~~ ME HAVE THE POSITIONS FOR THE 'MIRRORS

BEST REGARDS,

Andy.

Scale	1:200	Drawn By	SK001	Rev	(7)
Drawn As	14.2.9/04	Project			



PLEASE ADVISE
X, Y & Z FOR
CENTREPOINTS OF
POSSIBLE MIRROR
POSITIONS